UNIT – 3
CNS Pharmacology

Pharmacology of neurodegenerative disorders

Neurodegenerative diseases

- Alzheimer's disease (AD)
- Parkinson's disease (PD)
- Multiple sclerosis (MS)
- Amyotrophic lateral sclerosis (ALS)

Parkinson's disease (PD)



Dr. James Parkinson

(*An essay on the shaking palsy*)

Introduction

* A long term degenerative disorder of the CNS that mainly affects the motor system.
* A neurological movement disorder.
* The symptoms usually emerge slowly, and as the disease worsens, non-motor symptoms become more common.

Epidemiology

* Common in elderly (>60) (1 in 100 affected)
* Common in males (3:2 ratio)
* Based on a study conducted in 2015
  - 6.2 million people affected/ 117,400 died
* Life expectancy – between 7 and 15 years

Etiology

* Exact cause – unknown

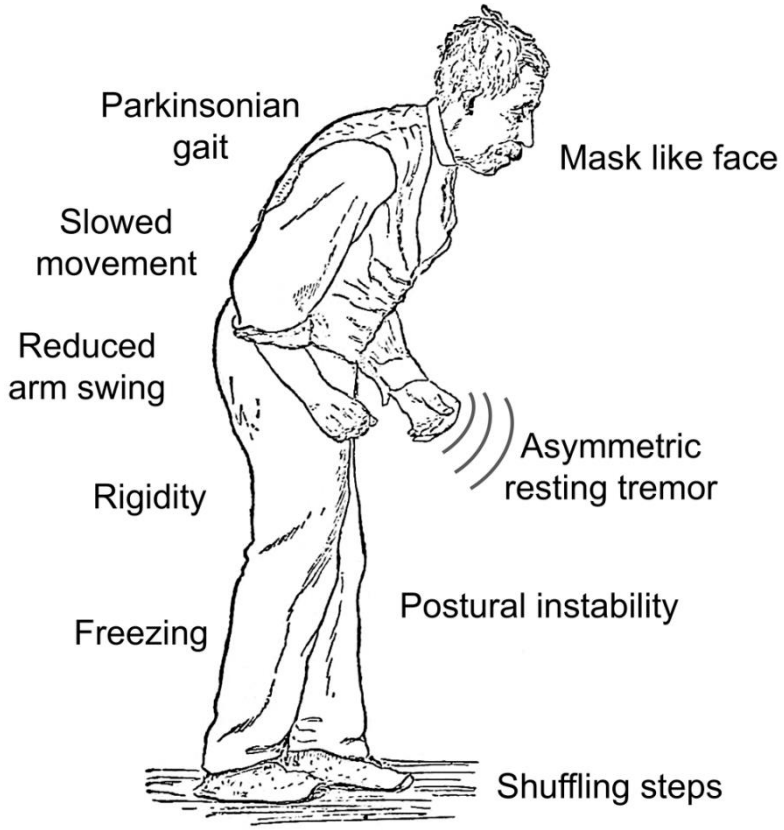
* Risk factors
  - Inherited
    * Genetic
  - Environmental
    * Protective – coffee, smoking , NSAIDs
    * Pathologic – pesticides, head injury

Sign and symptoms

* The 4 Cardinal symptoms of parkinsonism
  - Tremor
  - Bradykinesia
  - Rigidity
  - Postural instability

- Disorders of cognition , mood, behavior and thought follow with ANS alterations.

Motor symptoms



Pathophysiology

* PD occurs when nerve cells (neurons) in an area of the brain called the substantia nigra become impaired or die.

* These cells normally produce dopamine, a neurotransmitter that helps the cells of the brain communicate (transmits signals, “messages,” between areas in the brain).

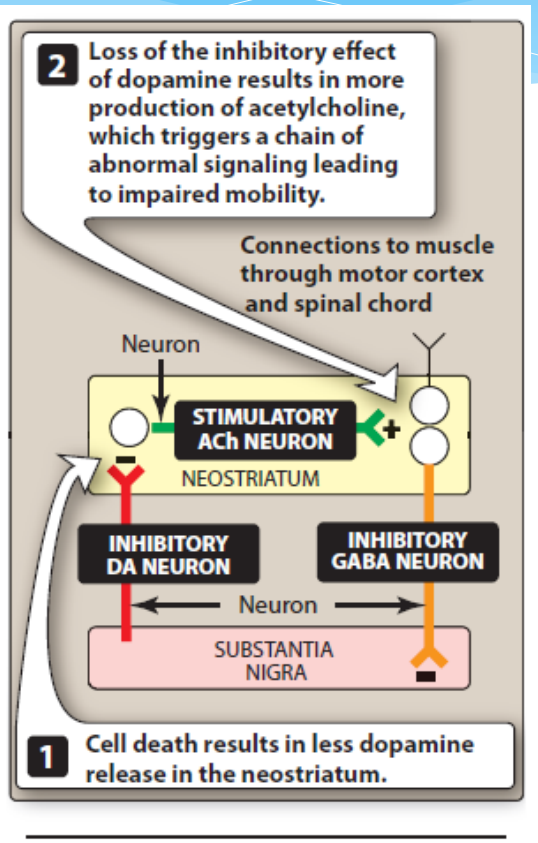
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* When these nerve cells become impaired or die, they produce less dopamine.
* Dopamine is especially important for the operation of basal ganglia.
* Basal ganglia is responsible for organizing the brain's commands for body movement.

Cont'd…

CNS neurotransmitters

- Inhibitory NTs
  - Dopamine
  - GABA
  - Glycine
- Excitatory NTs
  - Acetylcholine
  - Glutamate



Diagnosis

* Mainly based on clinical presentation
  * Parkinsonism
  * Pseudo parkinsonism

* Neuroimaging
  - MRI (Magnetic Resonance Imaging)
  - PET with DaT Scan (for differential diagnosis)

Treatment

* PD is incurable disease
* Goal of treatment
  * To reduce the effects of the symptoms
* Treatment modality
  * Initial therapy is promising
  * With disease progression, treatment becomes ineffective and side effect dominates
* Others (surgery, DBS)

Cont'd…

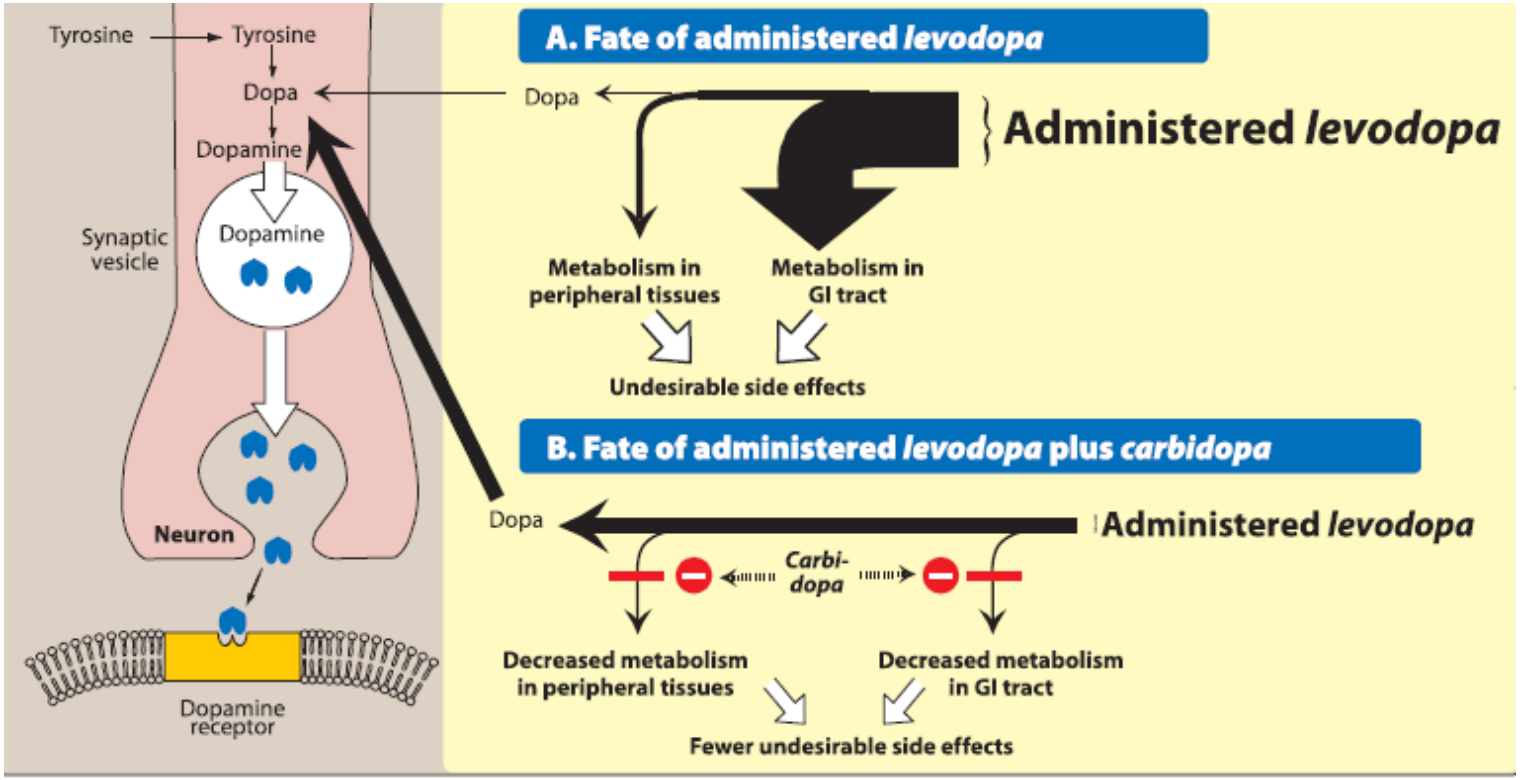
* Many currently available drugs aim to
  * Maintain CNS dopamine levels
  * Maintain signaling, as constant as possible.

* These agents offer temporary relief from the symptoms of the disorder, but they do not arrest or reverse the neuronal degeneration caused by the disease.

Drug groups

1. Levodopa and/or carbidopa
2. MAO-B inhibitors
3. COMT inhibitors
4. Dopamine receptor agonists
5. Anti-muscarinic drugs
6. Amantadine

1. Levodopa and carbidopa



Therapeutic uses

* Levodopa – carbidopa
  * Decreases rigidity, tremors, and other symptoms of parkinsonism.
  * Reduces the severity of symptoms for the first few years of treatment (in 2/3rd of patients).
  * Patients experience a decline in response during the 3rd to 5th year of therapy.
  * Withdrawal from the drug must be gradual.

Pharmacokinetics

* Absorbed rapidly from the small intestine.
* Extremely short half-life (1 to 2 hours)
* High protein meals interfere with the transport of levodopa into the CNS.

ADR and drug interactions

- ADR
  - Peripheral – anorexia, nausea, vomiting, hypo tension, tachycardia, brown urine and saliva
  - CNS effects – hallucination (both), dyskinesia, depression, anxiety, psychosis

- Drug Interactions
  - Pyridoxine (B6) – ↑peripheral breakdown
  - MAOI (non selective) – hypertensive crisis

2. MAO-B Inhibitors

* Drugs include selegiline and rasagiline
* MOA - selectively inhibits monoamine oxidase (MAO) type B, the enzyme that metabolizes dopamine.
* *Selegiline* co-administration with *levodopa*
  - Enhances the actions of *levodopa*
  - Substantially reduces the required dose.

3. COMT – Inhibitors

* Drugs include entacapone and tolcapone
* MOA – selectively and reversibly inhibit COMT and reduce methylation of levodopa.

Cont'd…

PK parameters	Data
Absorption	Unaffected by food
Distribution	Highly plasma protein bound
Metabolism	Hepatic and renal
Excretion	Urine/feces
Side effect	Same as carbidopa-levodopa
DOA	Tolcapone (Long DOA)

4.Dopamine receptor agonists

* Includes bromocriptine, pramipexole and apomorphine.
* Longer duration of action than *levodopa* and are effective in patients exhibiting fluctuations in response to *levodopa*.
* Apomorphine is injectable (used in advanced and severe stages to supplement PO drugs)

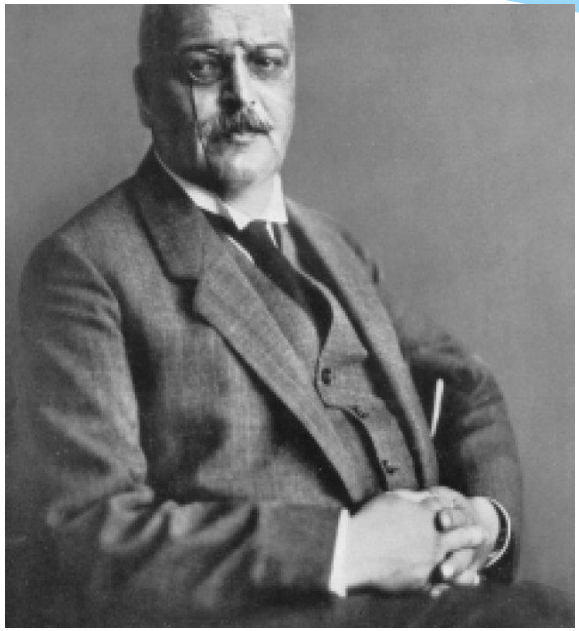
5. Anti-muscarinic drugs

* Includes benztropine and trihexyphenidyl
* Much less efficacious than *levodopa* and play only an adjuvant role in antiparkinsonism therapy.
* Induce mood changes & confusion, and produce xerostomia, constipation & visual problems.

6. Amantadine

* Antiviral agent discovered accidentally
* MOA – increases the release of dopamine, blocks cholinergic receptors, and inhibits the N-methyl-d-aspartate (NMDA) type of glutamate receptors.
* Less efficacious than levodopa and yet has fewer adverse effects.

Alzheimer's disease (PD)



Dr. Alois Alzheimer

(Psychiatrist and pathologist)

Introduction

* A long term degenerative disorder that starts slowly and progressively worsens.
* It is the cause of 60–70% of cases of dementia.
* The most common early symptom is difficulty in remembering recent events.

Epidemiology

* Common in elderly (>65) (6 in 100 affected)
* Common in females
* Based on a study conducted in 2015
  - 50 million people with dementia/AD seen in 60%
* Life expectancy – between 3 and 9 years

Etiology

* Exact cause – unknown

* Risk factors
- Inherited
  * Genetic
- Environmental factors
  * Hypertension, head injury , clinical depression

Sign and symptoms

* The 3 stages of Alzheimer's disease are
  - Early / mild
  - Middle / moderate
  - Late / severe

- AD is known to target the hippocampus which is associated with memory.

Cont'd…

* Early
  * Difficulties with language, executive functions, old memories are intact
* Middle
  * Long term memory impaired, patient cannot identify close relatives, can't read/write well
* Late
  * Language is reduced to simple phrases or even single words, eventually leading to complete loss of speech.

Hypotheses of AD

* Cholinergic hypothesis
  * AD is caused by reduced synthesis of the neuro transmitter acetylcholine.
* Amyloid hypothesis
* Tau hypothesis
* Inflammatory hypothesis
* cholesterol signaling hypothesis
* Neurovascular hypothesis

Diagnosis

* Clinical presentation
* Cognitive testing
* Neuroimaging
* Examination of brain tissue

Treatment strategies

* Pharmaceutical
  * Acetylcholinesterase inhibitors
  * NMDA receptor antagonists
* Psychosocial

* Caregiving

Cont'd…

* Current therapies aim to
  * Improve cholinergic transmission within the CNS
  * Prevent excitotoxic actions resulting from overstimulation of NMDA-glutamate receptors.

Cont'd…

* Pharmacologic intervention for Alzheimer disease is only palliative and provides modest short-term benefit.

* None of the available therapeutic agents alter the underlying neurodegenerative process.

AchE inhibitors

* Medications approved for the treatment of AD include *donepezil*, *galantamine* & *rivastigmine*.

* Have some selectivity for AChE in the CNS, as compared to the periphery.

NMDA receptor antagonist

* Memantine acts by blocking the NMDA receptor and limiting Ca2+ influx into the neuron, such that toxic intracellular levels are not achieved.
* Well tolerated, with few dose-dependent adverse events.
* Given in combination with an AChE inhibitor.

THE END

THANK YOU